



Comparative Analysis of ^{137}Cs Radiation Attenuation in Aluminum Using MCNP6 Simulations: Influence of Beam Geometry

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1. Introduction

The interaction of ionizing radiation with matter constitutes a fundamental basis for several applications in nuclear engineering, radiation shielding, medical physics and industrial process control [1]. When a monoenergetic beam of uncharged particles, such as photons, is normally incident on an attenuating medium of length x , the probability of interaction per unit path length is characterized by the linear attenuation coefficient (μ). Consequently, the radiation intensity decreases exponentially according to the Beer-Lambert law [2], depending on absorber thickness and the linear attenuation coefficient, which depends on the material composition, density, and photon interaction cross sections. The Monte Carlo method relies on the statistical sampling of particle histories to simulate radiation transport and optimize experimental geometries while reducing uncertainties associated with source-detector misalignment, scattered radiation [3]. This study evaluates the effluence of pencil-beam and divergent-beam geometries on the determination of the linear and mass attenuation coefficients of aluminum when subjected to 0.662 MeV gamma photons from a ^{137}Cs point source. The simulated results are compared directly against the standard reference database from the National Institute of Standards and Technology (NIST/XCOM) to quantify the influence of beam geometry on the accuracy of the calculated linear attenuation coefficients [4].

2. Methodology

The computational modeling of the attenuation experiment was developed within the Monte Carlo N-Particle (MCNP6) code with the aid of the MCNP Visual Editor (Vised) tool for the construction and verification of the three-dimensional geometry. The simulation setup arrangement consisted of four main components: a ^{137}Cs point source, an aluminum absorber, a NaI(Tl) detector, and a lead collimator with a diameter of 0.5 cm positioned directly in front of the detector to reduce the contribution of scattered photons. This post-target collimation was designed to reduce the contribution of scattered photons reaching the detector. The absorber consisted of a solid aluminum cylinder with a density of 2.6989 g.cm⁻³ positioned directly in front of the source. The absorber thickness, defined as the variable x , varied from 1 to 5 cm in increments of 1 cm for each simulation. Fig. 1 presents the three-dimensional geometry configuration employed in the attenuation simulations.

Two beam geometries were investigated. The first assembly employed a pencil beam, in which the radiation was emitted as an ideal, monodirectional beam aimed directed toward the center of the detector. The second configuration utilized a divergent beam modeled to cover the detector radius, establishing an initial semi-angle of 12.92°. This aperture was systematically reduced by half in successive simulations until reaching a minimum value of 1.62°, reproducing the behavior of practical collimation systems.

A reduction in the beam semi-angle decreases the angular divergence of the incident photons, approximating the narrow-beam condition assumed in ideal attenuation models. Consequently, smaller semi-angles reduce the probability of scattered photons reaching the detector and minimize geometric broadening effects, improving agreement with theoretical attenuation coefficients.

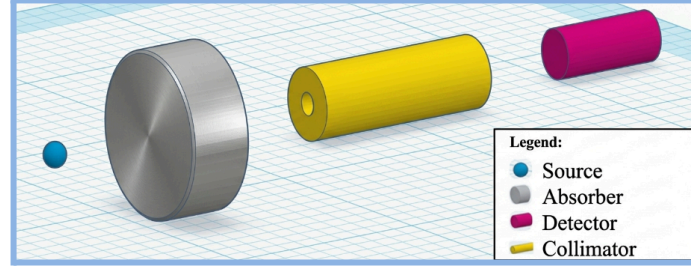


Figure 1: Three-dimensional geometry employed in the MCNP6 attenuation simulations.

For each absorber thickness and beam geometry configuration, the transmitted intensity values (I) and the initial intensity measured in the absence of the absorber (I_0) were extracted. The linear attenuation coefficient (μ) was calculated by rearranging the Beer-Lambert using the Eq. (1), expressed as

$$I = I_0 \cdot e^{-\mu x} \quad (1)$$

Where:

- I_0 : initial radiation intensity;
- μ : linear attenuation coefficient, representing the probability of photon interaction per unit length (cm^{-1});
- x : thickness of the sample (cm).

To approximate realistic experimental conditions, a scintillator detector based on a thallium-activated sodium iodide crystal, NaI(Tl), was modeled. The detector assembly included magnesium oxide (MgO) reflectors and an aluminum region employed as a simplified representation of the photomultiplier tube (PMT) with a thickness of 3 cm, as illustrated in Fig. 2 [5].

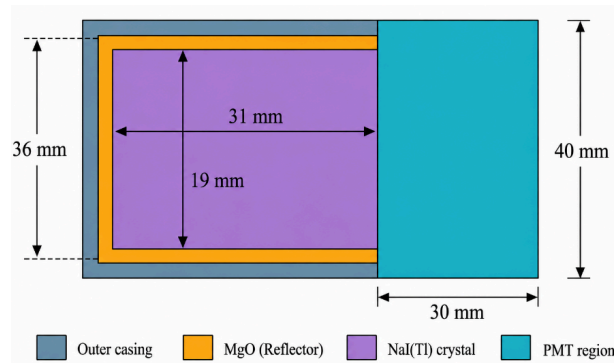


Figure 2: Cross-sectional view of the simulated NaI(Tl) detector modeled in MCNP6.

3. Results and Discussion

The experimental implementation was structured by varying the absorber's thickness and the divergent beam semi-angle. The aluminum attenuating medium was modeled with thicknesses ranging from 1.0 to 5.0 cm. For each length x , five beam geometries were simulated to evaluate the geometric dependence of the attenuation measurements. The baseline configuration employed a divergent beam

with a semi-angle of 12.92° , while the remaining geometries were generated by successively halving the angular aperture, characterizing the transition from a broad divergent beam toward the mathematical ideal of a pencil beam. The progressive reduction of the divergent beam semi-angle produced a systematic decrease in the relative error, demonstrating convergence toward the ideal narrow-beam attenuation condition represented by the pencil beam model.

To determine the linear attenuation coefficient, a linear regression analysis was performed on the $\ln(I_0/I)$ versus thickness x data, derived from Equation (1). The slope of the resulting linear fit directly yielded the value of μ for each configuration, as illustrated in Fig. 3:

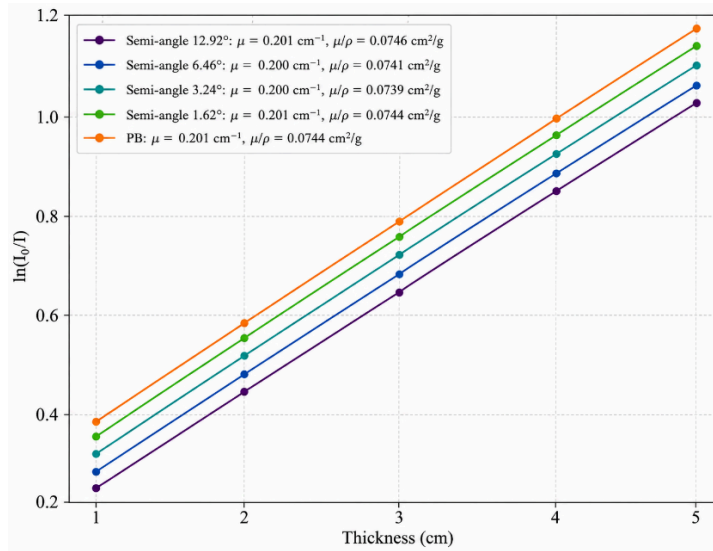


Figure 3: Comparison of attenuation linear regressions obtained for different beam geometries.

Based on the results derived from these linear regressions, Table I was compiled to present a comparison between the simulated values for the ^{137}Cs source (0.662 MeV) and the official reference data from the NIST/XCOM database, which provides a theoretical mass attenuation coefficient (μ/ρ) value of $7.433 \times 10^{-2} \text{ cm}^2 \cdot \text{g}^{-1}$ for aluminum.

Table I: Comparison of attenuation coefficients obtained for different beam semi-angles.

Beam Geometry	Semi-angle ($^\circ$)	μ (cm^{-1})	μ/ρ ($\text{cm}^2 \cdot \text{g}^{-1}$)	Relative Error (%)
Divergent Beam	12.92	0.201392	0.074620	0.39
	6.46	0.199867	0.074055	0.37
	3.24	0.199562	0.073942	0.52
	1.62	0.200894	0.074435	0.14
Pencil Beam	0	0.200673	0.074354	0.03

The pencil beam model demonstrated the highest accuracy, yielding a relative error of only 0.03% compared to NIST reference data. In contrast, the divergent geometries exhibited larger deviations, which are primarily attributed to their inherent geometric divergence. It should be noted that minor fluctuations in relative error, such as the value observed at 3.24° , are inherent to the statistical nature of Monte Carlo simulations. While the lead collimator effectively filters Compton-scattered photons, thereby ensuring the detector records only uncollided primary radiation, it cannot compensate for the geometric deviation of a divergent beam. Unlike the pencil beam, which represents a mathematically ideal parallel trajectory, the divergent beam causes photons to traverse an effective path length within the absorber slightly greater than the nominal thickness x . Consequently, this geometric discrepancy prevents the collimated configurations from achieving the same precision as the pencil beam model

when benchmarked against the idealized standards of the NIST/XCOM database. Additionally, divergent beam geometries increase the probability of photons undergoing Compton scattering before reaching the detector. Even with post-target collimation, secondary scattered photons generated within the absorber may contribute indirectly to the detected intensity, producing deviations from the ideal exponential attenuation behavior assumed in narrow-beam theoretical models such as NIST/XCOM.

4. Conclusions

The results demonstrated that beam geometry significantly influences the determination of attenuation coefficients in Monte Carlo simulations. The pencil beam configuration presented the best agreement with the NIST/XCOM reference data, yielding the lowest relative error among all evaluated configurations. In contrast, divergent beam geometries produced progressively larger deviations due to geometric divergence effects and increased contributions from scattered radiation. The progressive reduction of the beam semi-angle resulted in improved agreement with theoretical attenuation models, demonstrating convergence toward the ideal narrow-beam condition. These results highlight the importance of beam collimation and geometric control for accurate attenuation measurements and reinforce the influence of angular beam divergence on photon transport simulations.

Additionally, the study demonstrated the capability of MCNP6 simulations as a reliable virtual laboratory for the optimization and validation of radiation attenuation experiments, providing a cost-effective and safe alternative to physical experimental setups by reducing operational costs and radiation exposure risks.

As a future perspective, the proposed methodology may be extended to other attenuating materials and photon energies to evaluate the influence of beam geometry under different interaction regimes and experimental conditions.

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